

Embedded and IOT Systems



E&ICT Academy, IIT Kanpur
A Joint Initiative of MeitY & IIT Kanpur

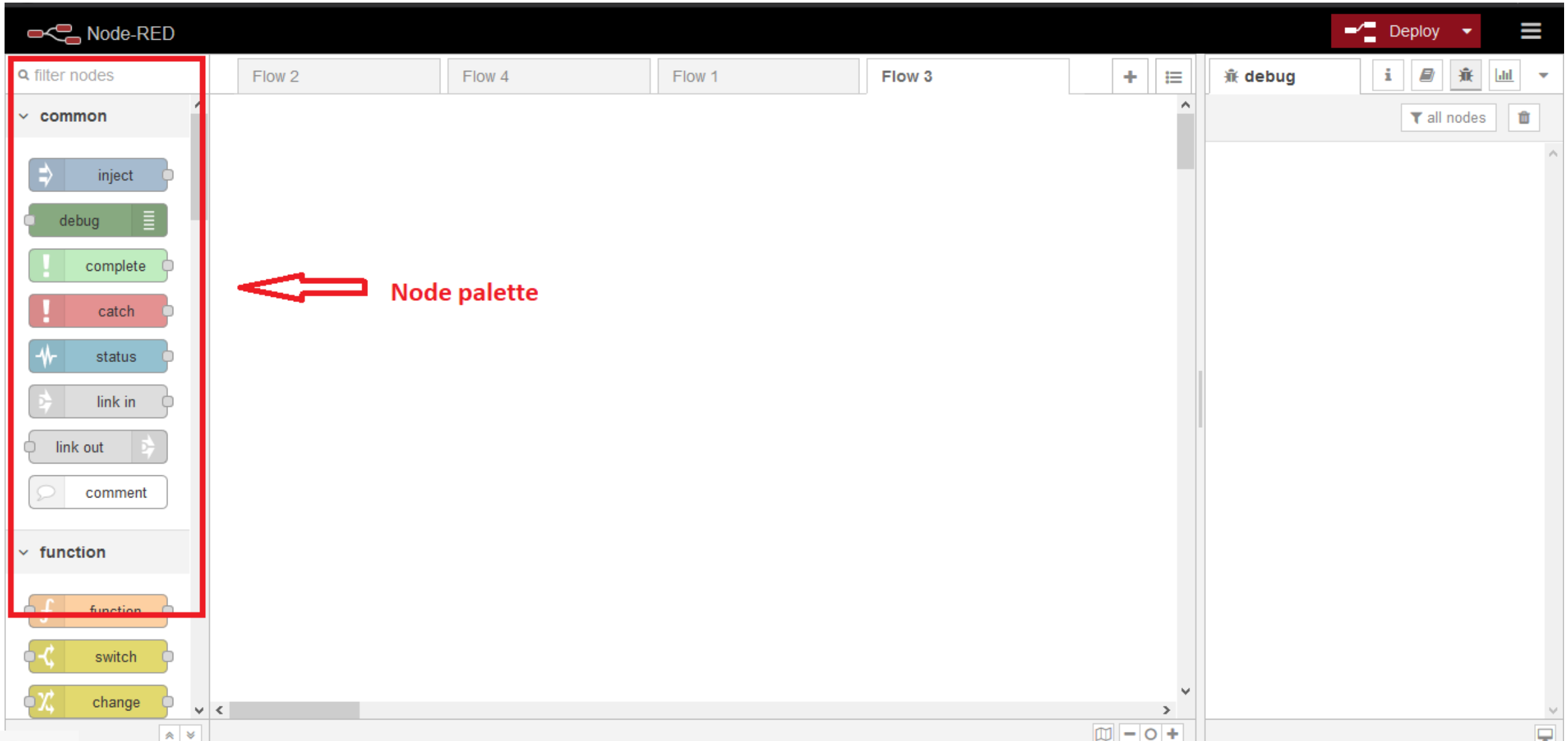
Today's Agenda

- **Flow in Node-RED**
- **Inject node**
- **Debug node**
- **Function node**
- **Dashboard SETUP**

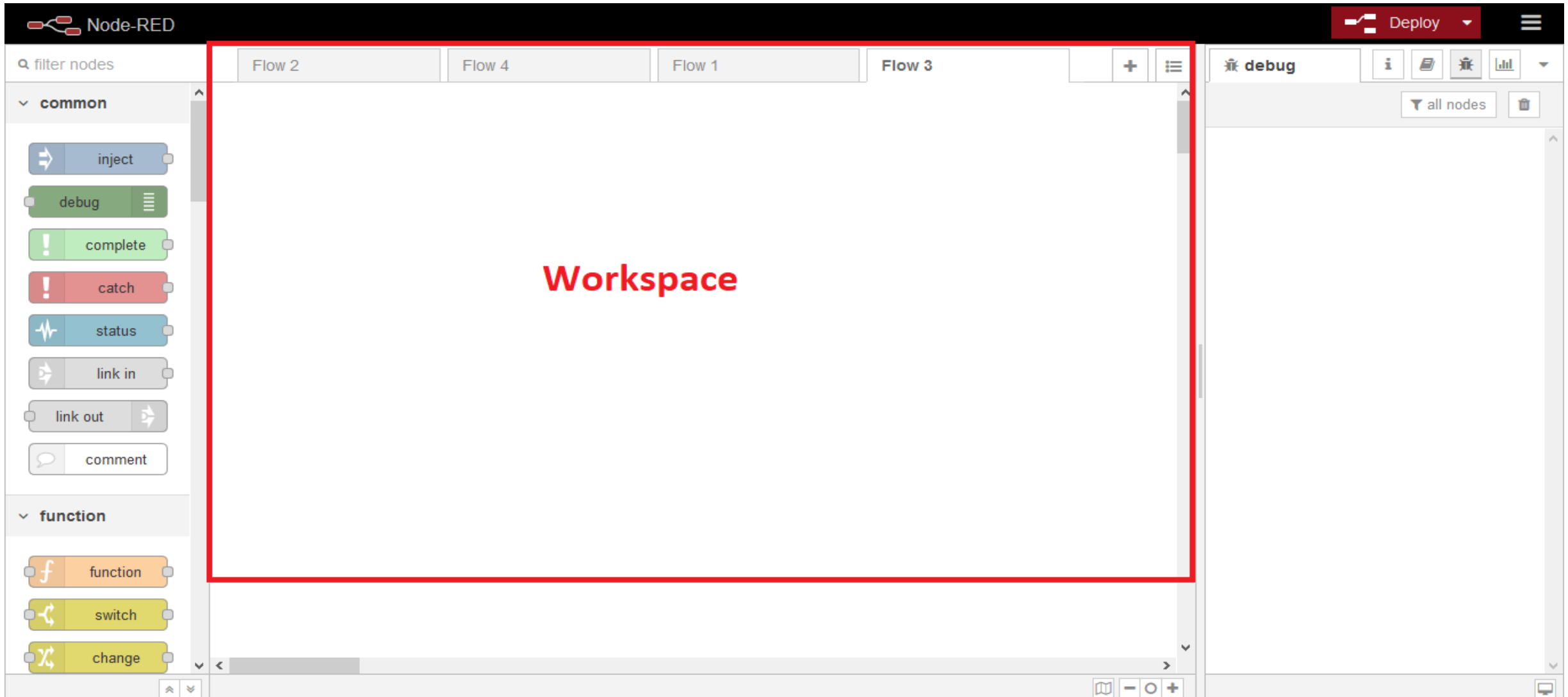
Main landing page

The screenshot displays the Node-RED web interface. At the top, a black header bar contains the Node-RED logo on the left and a red 'Deploy' button with a dropdown arrow on the right. Below the header, a light gray bar shows four flow tabs: 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. To the right of these tabs are icons for adding a new flow (+), a list view (three horizontal lines), and a 'debug' panel. The left sidebar is divided into two sections: 'common' and 'function'. The 'common' section includes nodes like 'inject', 'debug', 'complete', 'catch', 'status', 'link in', 'link out', and 'comment'. The 'function' section includes 'function', 'switch', and 'change' nodes. The main workspace is a large white area for building flows. On the right, the 'debug' panel is open, showing a search bar with 'all nodes' and a trash icon, and a large empty area for displaying node data.

Node palette (Left side)



Workspace (In Middle)



output pane(In right)

The image shows the Node-RED web interface. On the left is a palette of nodes categorized into 'common' (inject, debug, complete, catch, status, link in, link out, comment) and 'function' (function, switch, change). The central workspace contains four tabs labeled 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. On the right side, the 'debug' pane is open, displaying the text 'Output Pane' in red. The top of the interface features a 'Deploy' button and a hamburger menu icon. The bottom of the interface includes a status bar with zoom controls and a help icon.

Info Tab(in output pane)

The screenshot displays the Node-RED web interface. The top bar includes the Node-RED logo, a 'Deploy' button, and a menu icon. Below the top bar, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The main workspace contains a flow with two nodes: 'timestamp' and 'msg.payload'. The right-hand pane is open to the 'info' tab, which is highlighted with a red box. This pane shows a list of flows (Flow 2, Flow 4, Flow 1, Flow 3) and subflows, also highlighted with a red box. Below this list, the text 'info about flows' is displayed in green. At the bottom of the right pane, a detailed view of the selected 'timestamp' node is shown, including its ID ('39655d1f.997d62') and type ('inject').

Node-RED

Deploy

filter nodes

Flow 2 Flow 4 Flow 1 Flow 3

common

- inject
- debug
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

timestamp

msg.payload

info

Search flows

Flows

- Flow 2
- Flow 4
- Flow 1
- Flow 3
- Subflows
- Global Configuration Nodes

info about flows

timestamp

Node	"39655d1f.997d62"
Type	inject

show more

Help Tab(in output pane)

The screenshot displays the Node-RED web interface. The top bar includes the Node-RED logo, a 'Deploy' button, and a hamburger menu. Below the top bar, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The left sidebar contains a 'filter nodes' search bar and two categories of nodes: 'common' (inject, debug, complete, catch, status, link in, link out, comment) and 'function' (function, switch, change). The main workspace shows a flow with a 'timestamp' node connected to a 'msg.payload' node. The right sidebar is the 'output pane', which is currently showing the 'help' tab. The 'help' tab is highlighted with a red box. Below the 'help' tab, there is a search bar and a list of nodes. The 'inject' node is selected and highlighted with a red box. Below the node list, the 'inject' node's documentation is displayed, including a description, 'Outputs' section, and 'Details' section.

Node-RED

Deploy

filter nodes

Flow 2 Flow 4 Flow 1 Flow 3

common

- inject
- debug
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

timestamp

msg.payload

help

Search help

- http proxy
- http request
- http response
- inject
- join
- join

inject

Injects a message into a flow either manually or at regular intervals. The message payload can be a variety of types, including strings, JavaScript objects or the current time.

Outputs

payload *various*

The configured payload of the message.

topic *string*

An optional property that can be configured in the node.

Details

The Inject node can initiate a flow with a specific payload value. The default payload

Debug Tab(in output pane)

The screenshot displays the Node-RED web interface. At the top, the 'Node-RED' header is visible on the left, and a 'Deploy' button is on the right. Below the header, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The 'Flow 3' tab is currently selected. On the left side, a sidebar contains a search bar labeled 'filter nodes' and two categories of nodes: 'common' and 'function'. The 'common' category includes nodes like 'inject', 'debug', 'complete', 'catch', 'status', 'link in', 'link out', and 'comment'. The 'function' category includes 'function', 'switch', and 'change'. In the main workspace, there are two nodes: a 'timestamp' node and a 'msg.payload' node. On the right side, the 'output pane' is visible. It has a tab labeled 'debug' which is highlighted with a red rectangle. Other tabs in the output pane include 'all nodes' and a trash icon. The 'debug' tab is currently active, showing a list of debug messages.

Dashboard Tab(in output pane)

The screenshot displays the Node-RED web interface. At the top, the Node-RED logo and a 'Deploy' button are visible. Below the header, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The main workspace contains a flow with two nodes: a 'timestamp' node and a 'msg.payload' node. On the left, a sidebar lists various nodes under 'common' and 'function' categories. On the right, the 'Dashboard' tab is selected and highlighted with a red box. The dashboard configuration pane shows a 'Layout' section with 'Site' and 'Theme' options, and a 'Tabs & Links' section with a tree view showing 'Tab 1' containing 'Group 1'. A message at the bottom of the dashboard pane states: 'There are 6 widgets not in a group. Click here to create the missing groups'.

Before deploy(connect nodes)

The screenshot shows the Node-RED web interface. The top bar is black with the Node-RED logo and a 'Deploy' button highlighted with a red rectangle. Below the top bar, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The main workspace shows a flow with two nodes: a 'timestamp' node (blue) connected to a 'msg.payload' node (green). The left sidebar contains a search bar and two categories of nodes: 'common' and 'function'. The right sidebar shows the 'dashboard' settings, including 'Layout', 'Site', 'Theme', and 'Tabs & Links'. A message at the bottom of the right sidebar states: 'There are 6 widgets not in a group. Click here to create the missing groups'.

After deploy

The image shows the Node-RED web interface after a deployment. The top bar includes the Node-RED logo and a 'Deploy' button highlighted with a red rectangle. Below the top bar, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The 'Flow 3' tab is active, showing a flow with two nodes: a 'timestamp' node (blue) and a 'msg.payload' node (green). The 'timestamp' node is connected to the 'msg.payload' node. On the left side, there is a sidebar with a search bar 'filter nodes' and two categories of nodes: 'common' and 'function'. The 'common' category includes nodes like 'inject', 'debug', 'complete', 'catch', 'status', 'link in', 'link out', and 'comment'. The 'function' category includes 'function', 'switch', and 'change'. On the right side, there is a 'debug' tab and a search bar 'all nodes'.

PRESS INJECT in Workspace

The screenshot shows the Node-RED web interface. On the left, the 'common' category of nodes is expanded, showing 'inject', 'debug', 'complete', 'catch', 'status', 'link in', 'link out', and 'comment'. The main workspace contains a flow with an 'inject' node (highlighted by a red box and the text 'press it' with a red arrow), followed by a 'timestamp' node and a 'msg.payload' node. The right sidebar shows the 'debug' console with 'all nodes' selected. The top bar includes the 'Node-RED' logo, a 'Deploy' button, and a menu icon.

Miliseconds is showing in output pane

The image shows the Node-RED web interface. On the left, the 'common' nodes palette is visible, containing nodes like inject, debug, complete, catch, status, link in, link out, and comment. The main workspace shows a flow with a 'timestamp' node connected to a 'msg.payload' node. On the right, the 'debug' console is open, displaying a message from node 368de4b.4b57f9c: 'msg.payload : number' followed by the value '1598349635020'. A red box highlights this output, and a red arrow points down to the text 'number of miliseconds elapsed since 1st january 1970.'.

Node-RED

Flow 2 Flow 4 Flow 1 Flow 3

filter nodes

common

- inject
- debug
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

timestamp

msg.payload

debug

all nodes

8/25/2020, 3:30:35 PM node: 368de4b.4b57f9c
msg.payload : number
1598349635020

number of miliseconds elapsed since 1st january 1970.

CLICK ON MILLISECONDS

The screenshot displays the Node-RED web interface. On the left, the 'common' nodes palette is visible, containing nodes like inject, debug, complete, catch, status, link in, link out, and comment. The central workspace shows a flow with a 'timestamp' node connected to a 'msg.payload' node. On the right, the 'debug' console is open, showing a log entry for the node '368de4b.4b57f9c' with the message 'msg.payload : number' and the value '1598349635020'. A red box highlights the value '1598349635020', and a red arrow points to it with the text 'click it'.

Node-RED

Deploy

filter nodes

Flow 2 Flow 4 Flow 1 Flow 3

common

- inject
- debug
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

timestamp

msg.payload

debug

all nodes

8/25/2020, 3:30:35 PM node: 368de4b.4b57f9c

msg.payload : number

1598349635020

click it

ISO Representation

The image shows the Node-RED web interface. On the left, the 'common' nodes palette is visible, containing nodes like inject, debug, complete, catch, status, link in, link out, and comment. The 'function' nodes palette is also visible, containing function, switch, and change nodes. In the center workspace, a flow is built with a 'timestamp' node connected to a 'msg.payload' node. On the right, the 'debug' console is open, showing a log entry for the node '368de4b.4b57f9c'. The log entry displays 'msg.payload : number' and the ISO timestamp '2020-08-25T10:00:35.020Z', which is highlighted with a red box. A red arrow points from the text 'ISO representation' below the box to the timestamp string in the log.

Node-RED

filter nodes

Flow 2 Flow 4 Flow 1 Flow 3

common

- inject
- debug
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

timestamp

msg.payload

debug

all nodes

8/25/2020, 3:30:35 PM node: 368de4b.4b57f9c

msg.payload : number

2020-08-25T10:00:35.020Z

ISO representation

Click Again On ISO Representation

The image shows the Node-RED web interface. On the left, the 'common' node palette includes 'inject', 'debug', 'complete', 'catch', 'status', 'link in', 'link out', and 'comment'. The 'function' palette includes 'function', 'switch', and 'change'. The main workspace contains a flow with a 'timestamp' node connected to a 'msg.payload' node. On the right, the 'debug' console displays the message: '8/25/2020, 3:30:35 PM node: 368de4b.4b57f9c msg.payload : number 2020-08-25T10:00:35.020Z'. The ISO timestamp '2020-08-25T10:00:35.020Z' is highlighted with a red box, and a red arrow points to it with the text 'click again'.

UTC Representation of time

The image shows the Node-RED web interface. On the left, the 'common' nodes palette is visible, containing nodes like inject, debug, complete, catch, status, link in, link out, and comment. Below it, the 'function' palette contains function, switch, and change nodes. The main workspace shows a flow with a 'timestamp' node connected to a 'msg.payload' node. On the right, the 'debug' console displays the output of the flow. The output is a date and time in UTC, highlighted by a red box and an arrow pointing to it with the text 'DATE,time in UTC'.

Node-RED interface showing a flow with a timestamp node connected to a msg.payload node. The debug console displays the output: 8/25/2020, 3:30:35 PM [UTC+5.5].

DATE,time in UTC

CLICK AGAIN on UTC

The screenshot shows the Node-RED web interface. The top bar includes the Node-RED logo, a 'Deploy' button, and a hamburger menu. Below the top bar, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The left sidebar contains a search bar and two categories of nodes: 'common' and 'function'. The 'common' category includes nodes like 'inject', 'debug', 'complete', 'catch', 'status', 'link in', 'link out', and 'comment'. The 'function' category includes 'function', 'switch', and 'change'. The main workspace shows a flow with a 'timestamp' node connected to a 'msg.payload' node. The right sidebar shows the 'debug' console with a message log. The message log shows a timestamp '8/25/2020, 3:30:35 PM' and a node ID '368de4b.4b57f9c'. The message payload is 'number'. The timestamp is highlighted with a red box, and a red arrow points to it with the text 'CLICK AGAIN'.

Node-RED

filter nodes

Flow 2 Flow 4 Flow 1 Flow 3

common

- inject
- debug
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

timestamp

msg.payload

debug

all nodes

8/25/2020, 3:30:35 PM node: 368de4b.4b57f9c

msg.payload : number

8/25/2020, 3:30:35 PM [UTC+5.5]

CLICK AGAIN

127.0.0.1:1880/#5a1609ee.f3d0a8

HEXADECIMAL Representation of milli seconds

The image shows a Node-RED interface with a flow named 'Flow 3'. The flow consists of a 'timestamp' node connected to a 'msg.payload' node. The 'debug' console on the right shows the output of the flow, which is the hexadecimal value '0x174250fedcc'. The value is highlighted with a red box, and a red arrow points to it with the word 'HEXADECIMAL' written in red. The 'debug' console also shows the date and time '8/25/2020, 3:30:35 PM' and the node ID 'node: 368de4b.4b57f9c'.

Node-RED interface showing a flow with a timestamp node connected to a msg.payload node. The debug console displays the output: 8/25/2020, 3:30:35 PM node: 368de4b.4b57f9c msg.payload : number 0x174250fedcc. The value 0x174250fedcc is highlighted with a red box, and a red arrow points to it with the word HEXADECIMAL written in red.

(Dashboard setup)



Installation of Dashboard



Click on 3 Bars at upper right side

The screenshot displays the Node-RED web interface. At the top, the 'Node-RED' logo is on the left, and a 'Deploy' button is on the right. A red box highlights the three-bar menu icon in the top right corner. Below the header, there are tabs for 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The main workspace shows a flow with three nodes: a 'hello world!' node, a 'function' node, and a 'msg.payload' node. The left sidebar contains a 'filter nodes' search bar and a list of nodes including 'debug', 'complete', 'catch', 'status', 'link in', 'link out', 'comment', and a 'function' category with 'function', 'switch', 'change', 'range', and 'template' nodes. The right sidebar shows the 'debug' console with the output 'hello from Node-red!'.

Click On Manage Palette

The screenshot displays the Node-RED web interface. At the top, the 'Node-RED' logo is visible. Below it, a tab bar shows 'Flow 2', 'Flow 4', 'Flow 1', and 'Flow 3'. The main workspace contains a flow with three nodes: a 'hello world!' node, a 'function' node, and a 'msg.payload' node. On the left, a 'filter nodes' search bar is present, followed by a list of nodes including 'debug', 'complete', 'catch', 'status', 'link in', 'link out', 'comment', and a 'function' category with sub-nodes like 'function', 'switch', 'change', 'range', and 'template'. On the right, a 'Deploy' button and a menu icon are shown. The menu is open, displaying options: 'View', 'Import', 'Export', 'Search flows', 'Configuration nodes', 'Flows', 'Subflows', 'Groups', 'Manage palette' (highlighted with a red rectangle), 'Settings', 'Keyboard shortcuts', 'Node-RED website', and 'v1.1.3'.

CLICK ON INSTALL

The screenshot shows the Node-RED interface with the 'User Settings' dialog box open. The dialog has a 'Close' button in the top right corner. The 'View' tab is selected, showing a table with one row: 'Nodes' and 'Install'. The 'Install' button is highlighted with a red rectangle. Below the table, there is a 'Keyboard' section and a 'Palette' section. The 'Palette' section shows '2755 modules available'. The 'Nodes' section shows a search bar and a 'sort' dropdown menu with options 'a-z' and 'recent'. The 'Nodes' section also shows a 'search modules' input field with the text '2755' next to it. The 'Nodes' section also shows a 'sort' dropdown menu with options 'a-z' and 'recent'. The 'Nodes' section also shows a 'search modules' input field with the text '2755' next to it.

Node-RED

Deploy

filter nodes

Flow 2

debug

complete

catch

status

link in

link out

comment

function

function

switch

change

range

template

User Settings

Close

View

Nodes

Install

Keyboard

sort: a-z recent

search modules

2755

Palette

2755 modules available

debug

all nodes

8/25/2020, 4:05:52 PM node: 368de4b.4b57f9c

msg.payload : string[20]

"hello from Node-red!"

Install node-red-dashboard

The screenshot shows the Node-RED interface with the 'User Settings' dialog open. The 'Nodes' tab is selected, and the search bar contains 'node-red-dash'. The search results list several dashboard-related nodes. The first result, 'node-red-dashboard', is highlighted with a red box. A green arrow points to this result with the text 'install it'. The 'installed' button next to this node is highlighted in red. The 'function' palette is visible on the left, and the 'debug' console on the right shows a message: 'hello from Node-red!'.

Node-RED

filter nodes

Flow 2

debug

complete

catch

status

link in

link out

comment

function

function

switch

change

range

template

User Settings

View

Nodes

Install

Keyboard

sort: a-z recent

node-red-dash

6 / 2755

node-red-dashboard

A set of dashboard nodes for Node-RED

2.23.2 2 weeks ago

install

node-red-node-ui-list

Node-RED Dashboard UI widget node for simple list

0.3.1 2 months ago

install

node-red-node-ui-vega

Node-RED UI widget node for Vega visualization grammar

0.1.2 11 months ago

install

node-red-node-ui-table

Table UI widget node for Node-RED Dashboard

0.3.4 1 month ago

install

node-red-node-ui-microphone

debug

8/25/2020, 4:05:52 PM node: 368de4b.4b57f9c

msg.payload : string[20]

"hello from Node-red!"

npm Install node-red-dashboard

Command Prompt

```
Microsoft Windows [Version 10.0.17763.1098]  
(c) 2018 Microsoft Corporation. All rights reserved.  
  
C:\Users\Jitesh>npm install node-red-dashboard_
```

Check Dashboard on your Node-RED Installation

The screenshot displays the Node-RED web interface. On the left sidebar, the 'dashboard' category is expanded and highlighted with a red rectangle. It contains the following nodes: button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. The main workspace shows a flow with three nodes: a 'hello world!' node, a 'function' node, and a 'msg.payload' node. The right sidebar shows the 'debug' console with the following output:

```
8/25/2020, 4:05:52 PM node: 368de4b.4b57f9c  
msg.payload : string[20]  
"hello from Node-red!"
```